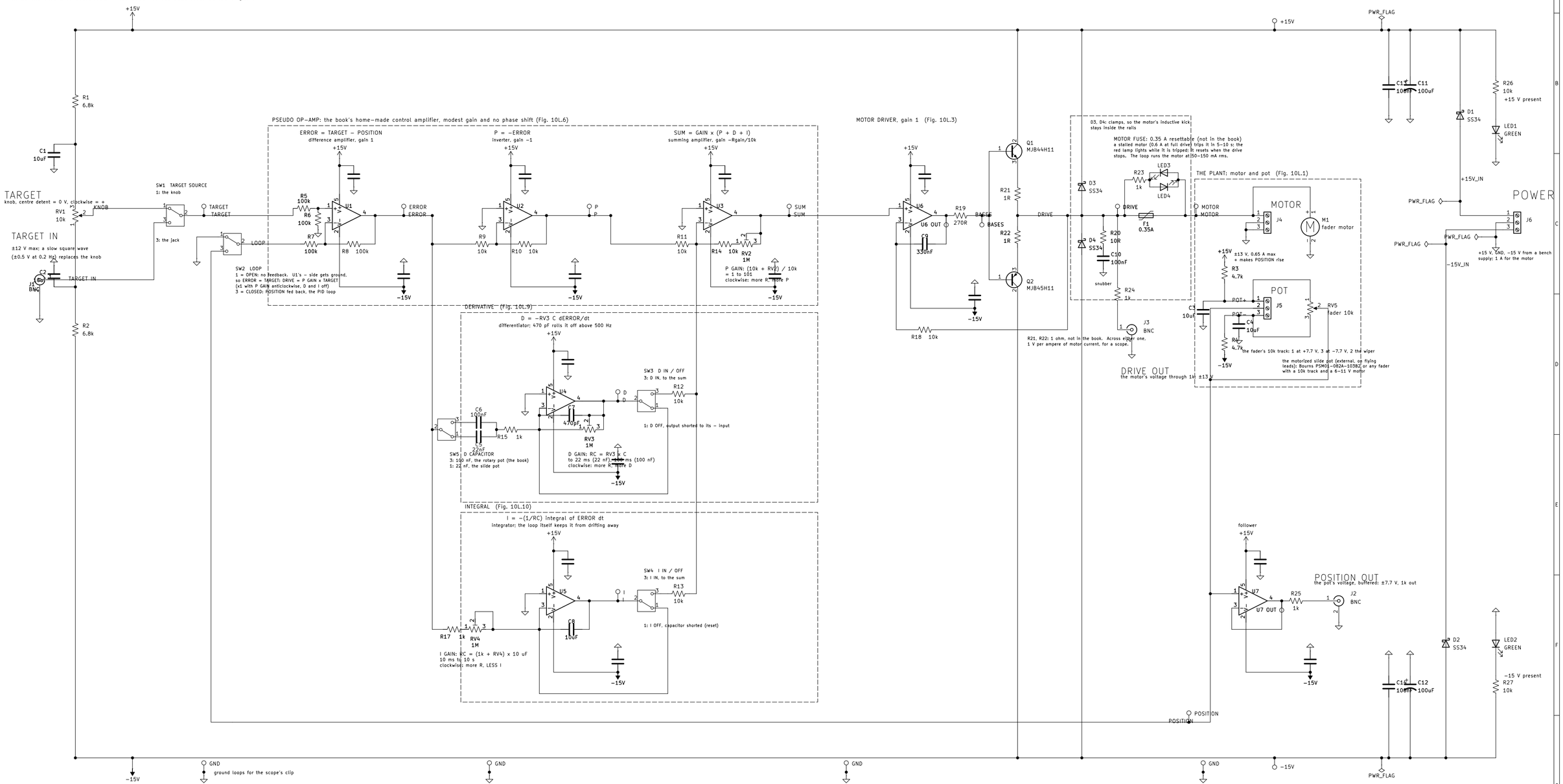
PID MOTOR CONTROL LOOP

Lab 10L of Learning the Art of Electronics (Hayes & Horowitz, 2016): a potentiometer sets a target position, a DC motor drives a second potentiometer to match it, and the controller is the book's home-made 'pseudo op-amp' of modest, adjustable gain, with the derivative and the integral of the error summed in. The board is the book's Fig. 10L.10 with every amplifier its own LM321 (the single LM358), the book's MJE3055/2955 as D2PAK MJB44H11/45H11, and its resistor substitution boxes as trimmers.

Why a home-made op-amp: the motor integrates (voltage in, position out: −90 degrees from DC), so an ordinary op-amp's own −90 degrees would make −180 at a few hertz. The pseudo op-amp has gain 1 to 101 and no phase shift of its own. Simulated as drawn (sim/pid_sim.py, the fader modelled from bench measurements): P only rings at 31 Hz when the gain nears 100; D equal to P there wants RC = 1 / 2 pi f = 5 ms (RV3 near 234k with 22 nF), and then gain 100 settles in 28 ms with 3 % overshoot. I removes the residual error at low P gain.

MH1 MH2 MH3 MH4

M3 mounting holes, one per corner




TARGET
Knob or jack, on the TARGET SOURCE switch. The knob's 6.8k ends keep it to ±6.4 V, inside what the motor pot can reach (±7.7 V): a target the loop can always meet. A ±3 V square wave at 0.2 Hz on the jack gives steps; a ±3 V triangle at 0.05 Hz, stick-slip. ±12 V max.

LOOP (SW2): which voltage U1 subtracts from TARGET. CLOSED (3): POSITION, the fader's wiper: the loop. OPEN (1): ground, so ERROR = TARGET and the motor driver gets P GAIN x TARGET, open loop: the knob or a generator drives the motor directly (x1 with P GAIN anticlockwise, D and I off). The slider runs to an end and stalls there, heating the driver: TARGET back to 0.

PSEUDO OP-AMP (U1, U2, U3)
U1 takes TARGET − POSITION with four 100k. U2 inverts it so P, D and I all arrive inverted at U3, which sums them through 10k each and multiplies by RV2's (10k + RV2)/10k = 1 to 101: the P GAIN. All three op-amps work at low closed-loop gain, so the chain has no phase shift where the loop's crossover is (a few hertz). At P gain 1.5 a 1 V step cannot unstick the slider; near 20 it overshoots; near 100 (P only) it rings: the period of that 'natural oscillation' sets D below.

Op-amps are drawn + on top: that is how the SOT-23-5 sits.

DERIVATIVE (U4, SW3, SW5)
D = −RV3 C dERROR/dt, rolled off above 500 Hz by the 470 pF. Set RV3 C to 1/6 of the natural period: for this slide pot (T about 32 ms) that is 5 ms, about 234k with the 22 nF: for the book's rotary motor-pot (T 0.6 s) use 100 nF and 1M. SW5 picks the capacitor. SW3 puts D into the sum or shorts it out.

INTEGRAL (U5, SW4)
I = −(1/RC) Integral ERROR dt, RC = (1k + RV4) x 10 uF. OFF shorts the capacitor, so I starts from zero. I drives the residual error to zero: too much I rings. Trimmers: clockwise is more resistance (Bourns: wiper to 3), so more P, more D, but LESS I. Arcs on the board.

MOTOR DRIVER (U6, Q1, Q2)
The book's Fig. 10L.3: the op-amp drives the push-pull through 270 ohm and the 10k takes the feedback from the motor side, so crossover distortion is inside the loop: the 0.33 uF feeds the op-amp's own output straight back so the phase-shifting transistors are bypassed at high frequency (split feedback, Lab 9L). 10 ohm + 0.1 uF snub the brush hash. Output ±13 V: the motor is rated 10 V, 6–11 V running, and sees full drive only in a step.

FUSE: not in the book. The 0.35 A PTC opens a full-drive stall (0.6 A) in seconds and a short in well under one, then resets itself. A stall at a few volts it holds, and one transistor takes up to 3.7 W, so they are D2PAKs on 2 oz copper pours. R21, R22 read the motor current, 1 V/A.

THE PLANT
The lab's Bourns PSM01-082A-103B2, or any fader with a 10k track and a 6–11 V motor. Bourns: motor A to MOTOR, B to GND; track 1, 2, 3 to POT 1, 2, 3 (A + moves the lever toward 1). Others: POT 1 to the end that must read +7.7 V; if the loop runs away, swap the motor wires. The servo track (1' 2' 3') and touch pin T stay free. POWER: ±15 V, 1 A (the motor's stall current). Series Schottkys against a reversed supply: 100 uF a rail for the motor's pulses; a green LED per rail. Quiescent about 10 mA. 100 nF at every op-amp pin.

Open source, CERN-OHL-P-2.0 -- github.com/gallicchio/motor-pid
P from 1 to 101, D up to 10 ms (slide pot) or 100 ms (rotary), I up to 10 s
LM321 (the single LM358) in SOT-23-5, one per triangle; MJB44H11 / MJB45H11 (D2PAK) push-pull
**Circuit: Hayes & Horowitz, Learning the Art of Electronics (2016), Lab 10L | Board: Jason Gallicchio**

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***Title: PID Motor Control Loop: Lab 10L of Learning the Art of Electronics***

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